

FOR METABRAINZ FOUNDATION · LEARNING SOFTWARE FOR CHILDREN

ListenBrainz — Advanced Interactive Visualizations

Google Summer of Code 2026 — Project Proposal

Medium Project · ~175 Hours · June 2 - August 25, 2026

👤 CONTRIBUTOR DETAILS	🎓 ACADEMIC DETAILS
Name Abhishek Verma	University JSS Academy of Technical Education, Noida
Email abhishekverma0700@gmail.com	Degree B.Tech Computer Science (2022-2026)
Phone +91 9452601372	CGPA 7.91 / 10.0
Location Lucknow, India (GMT+5:30)	Project ListenBrainz Advanced Visualizations
GitHub github.com/abhishekverma0700	Org MetaBrainz Foundation
Portfolio abhishek0700.netlify.app	Size Medium (~175 hours)
Projects portvia.vercel.app	Period June 2 – August 25, 2026

1 Time Commitment

Phase	Dates	Days	Hrs/Day	Est. Hours
Community Bonding	May 1 – June 1, 2026	32 days	4 hrs	~128 hrs
Coding Period 1	June 2 – July 14, 2026	43 days	6 hrs	~258 hrs
⚠ Exam Period	July 1 - July 7, 2026	7 days	3 hrs	~21 hrs
Coding Period 2	July 14 – Aug 25, 2026	43 days	6 hrs	~258 hrs
TOTAL	May 1 - Aug 25, 2026	~118 days	Avg 5.5 hrs	~350 hrs

No other major commitments during GSoC. Exam conflicts will be communicated to mentors at least one week in advance.

2 About Me

I am **Abhishek Verma**, a final-year B.Tech CS student at JSS Academy of Technical Education, Noida (CGPA: 7.91). I am a **full-stack developer and AI enthusiast** with 4 live production apps. I currently intern at Brixo Solutions building React Native apps in an agile team, and have solved **200+ LeetCode problems**.

Technical Skills

Category	Technologies
Frontend	React.js, React Native, JavaScript ES6+, HTML5, CSS3, Tailwind CSS
Backend	Node.js, Express.js, FastAPI, Django, REST API Design
AI & LLM	LangChain, RAG Pipelines, OpenAI API, Gemini, Prompt Engineering
Databases	MongoDB, MySQL, PostgreSQL, Redis, FAISS, ChromaDB
DevOps	Docker, AWS (EC2, S3), Firebase, GitHub Actions, CI/CD, Vercel

3 Professional Experience

Software Developer Intern — Brixo Solutions Pvt. Ltd., Lucknow

Aug 2024 - Present

React Native

Node.js

Firebase

REST APIs

Agile

- ▶ Built **25+ React Native screens** — billing, inventory, dashboard, and vehicle modules for end-to-end business operations.
- ▶ Implemented **OTP-based auth**, improved signup completion by **18%**.
- ▶ Optimized app performance via lazy loading and API caching — improved responsiveness by **30%**.
- ▶ Reviewed **20+ PRs** in 5-member agile team with shared design system.

4 Why MetaBrainz Foundation?

MetaBrainz Foundation's **Music Blocks** is a web-based, JavaScript-heavy educational platform — exactly the kind of project where my React.js and Node.js skills shine. The Git Version Control project directly aligns with my experience building full-stack apps with GitHub Actions CI/CD, REST APIs, and real-time data features. I have also built **DocuMind** — an AI-powered RAG system — making me uniquely qualified for the AI reflection component of this project. Contributing to software that teaches children worldwide is personally meaningful to me.

5 Project Goals & Deliverables

ListenBrainz currently has basic charts but lacks rich, interactive, personalized visualizations. I will build advanced data visualizations using Nivo and React that make music listening data meaningful and engaging:

Phase 1 — Core Visualization Suite (Weeks 1-6)

- ▶ **Interactive Listening Timeline:** Nivo-powered timeline chart showing listening history with zoom, pan, and drill-down by artist/album/genre.
- ▶ **Artist & Genre Heatmaps:** Calendar and radial heatmaps showing listening patterns by hour, day, and month.
- ▶ **Music Discovery Graph:** Force-directed network graph showing artist similarity and discovery paths using D3/Nivo.
- ▶ **Real-Time Dashboard:** Live stats panel with WebSocket updates for currently playing tracks and trending data.

| Phase 2 — Advanced Features & Personalization (Weeks 7-12)

- ▶ **Centralized Notification System:** Unified notification hub across all MetaBrainz sub-projects — replacing scattered standalone notifications.
- ▶ **Onboarding Experience:** Smooth, intuitive onboarding flow for new users with feature walkthroughs and trust-building UI.
- ▶ **Funkwhale/Navidrome Integration:** Allow users to play music from self-hosted servers directly in BrainzPlayer.
- ▶ **Test Coverage:** 80%+ Jest + React Testing Library coverage on all new visualization components.

| Stretch Goals

- ▶ Export visualizations as shareable images or embeddable widgets
- ▶ AI-powered listening insights using LangChain — "Why do you listen more on Fridays?"

6 Detailed Timeline

Phase	Week	Dates	Deliverables & Tasks
Community Bonding	W1-W4	May 1-June 1	Study ListenBrainz codebase and existing chart components. Discuss architecture with mentors. Set up dev environment. Write technical design doc for visualization suite.
Coding Period 1	W1	June 2-8	Set up Nivo library integration. Design component architecture for all new charts. Create feature branch.
	W2	June 9-15	Build interactive listening timeline — zoom, pan, drill-down by artist/album/genre. Unit tests.
	W3	June 16-22	Build artist and genre heatmaps — calendar view and radial charts. Responsive design.
	W4	June 23-29	Build music discovery network graph using D3/Nivo. Artist similarity and discovery paths.
	W5	June 30-July 6	Real-time dashboard with WebSocket updates. Live stats, currently playing, trending data.
	W6	July 7-13	Integration testing. Performance optimization. Accessibility audit. PR review with mentors.
Mid-Point Eval	—	July 14	✓ Timeline chart ✓ Heatmaps ✓ Network graph ✓ Real-time dashboard ✓ Tested & documented
Coding Period 2	W7	July 14-20	Centralized notification system for all MetaBrainz sub-projects. Notification hub UI.
	W8	July 21-27	Smooth onboarding flow for new users. Feature walkthroughs, trust-building UI components.
	W9	July 28-Aug 3	Funkwhale/Navidrome integration — play self-hosted music directly in BrainzPlayer.
	W10	Aug 4-10	Comprehensive Jest + React Testing Library coverage — 80%+ on all visualization components.
	W11	Aug 11-17	Full documentation — API docs, component storybook, contributor guide. Code cleanup. Final mentor review.
	W12	Aug 18-25	Final submission. GSoC report. Export/share feature. Post-GSoC contribution plan.
Final Eval	—	Aug 25	✓ Full viz suite ✓ Notifications ✓ Onboarding ✓ BrainzPlayer ✓ 80%+ tests ✓ Docs

7 Why I Stand Out



Data Visualization Skills

Built DocuMind with NumPy/Pandas data pipelines and Amala Bioceuticals with scroll animations and dynamic UI. I understand data-driven interfaces deeply — critical for building Nivo/D3 visualizations.



React.js Expert

ListenBrainz is a React-heavy platform. I have built 4 production React.js apps — Eduavaa, Amala Bioceuticals, Portvia, Abhirent. I understand React component architecture, state management, and performance optimization at a production level.



Real-Time Features

Built Abhirent with Firebase Realtime DB and real-time location sync. The ListenBrainz real-time dashboard requires exactly this — WebSocket and live data update experience I already have in production.



Production-Grade Code

4 live apps: eduavaa.me, amalabioceticals.in, abhirent.netlify.app, abhishek0700.netlify.app. I write code that ships to real users, handles real traffic, and stays maintainable long-term.



DevOps Ready

Docker, AWS, GitHub Actions CI/CD, Vercel. I can handle backend deployment and infrastructure independently — won't need mentor help for ops tasks, freeing up time for feature work.



Strong Fundamentals

200+ LeetCode problems solved. Agile team experience with PR reviews at Brixo Solutions. MERN Stack + Python/Django certifications. Strong base for tackling complex codebase challenges.

8 Important Links

Resource	URL
Portfolio	abhishek0700.netlify.app
Projects Showcase	portvia.vercel.app
GitHub	github.com/abhishekverma0700
LinkedIn	linkedin.com/in/abhishek-verma-27529221b
Live — Eduavaa (EdTech)	eduavaa.me
Live — Amala Bioceuticals	amalabioceuticals.in
Live — Abhirent	abhirent.netlify.app

Thank you for reviewing my proposal. I am genuinely excited to contribute to ListenBrainz and help music lovers worldwide explore their listening data through beautiful, interactive visualizations. My React.js expertise, real-time data experience, and passion for open-source music metadata make me an ideal fit for this project.

— **Abhishek Verma**